

Highlights

Adaptive Traffic Control in Urban Networks: A Physics-Informed Reinforcement Learning Approach

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- Identification of “empty road green light” inefficiencies in Hsinchu City’s traffic network.
- Modeling of a Barcelona-inspired *Eixample* grid as a robust testbed for traffic control.
- Implementation of a decentralized Max Pressure agent for real-time signal phase optimization.
- Comparative analysis showing the adaptive agent outperforms fixed-time control in high-density regimes.

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Abstract

Traffic congestion in Hsinchu City, particularly during weekday peak hours, is frequently exacerbated by inefficient signal timing. A common and frustrating phenomenon occurs when a congested arterial road faces a red light, while the intersecting road—despite being completely empty—retains a green light for an extended duration. This inefficiency in allocating green time contributes to gridlock and wasted capacity. To address this, we develop a simplified “toy model” for adaptive traffic control. We construct a grid-based road network inspired by the *Eixample* district of Barcelona, Spain, which serves as an ideal structural proxy for studying urban gridlocks due to its uniform, high-density layout. We simulate a $N \times N$ “Tian” grid using the physics-based Cell Transmission Model (CTM) and implement a decentralized Max Pressure control agent. Our results demonstrate that the adaptive agent, unlike fixed-time benchmarks, effectively eliminates “ghost green lights” by dynamically shifting priority to congested lanes, significantly reducing global density and average waiting times.

Keywords: Traffic Signal Control, Reinforcement Learning, Cell Transmission Model, Max Pressure, Barcelona Eixample

JEL: C63, R41, C61

1. Introduction

Urban traffic congestion is a critical issue in Hsinchu City, particularly during weekday commutes when high volumes of traffic overwhelm the existing infrastructure. A primary source of inefficiency in the current network is the lack of situational awareness in traffic signal controllers. It is a common occurrence for drivers on a jammed main artery to wait at a red light, while the opposing cross-street—having a green light—sits completely empty (Chen et al., 2025). This “ghost green light” phenomenon wastes valuable intersection capacity and exacerbates the frustration of commuters, leading to ripple effects that can paralyze the wider network.

The complexity of solving this problem lies in the non-linear dynamics of urban traffic, where local inefficiencies propagate as shockwaves. While modern Deep Reinforcement Learning (DRL) offers potential solutions (Liang et al., 2019), applying complex models directly to city-scale networks is computationally prohibitive and difficult to interpret. Therefore, a simplified yet structurally realistic “toy model” is required to isolate and solve the fundamental control problem before scaling up.

To this end, we draw inspiration from the *Eixample* district of Barcelona, Spain. Designed by Ildefons Cerdà in the 19th century, the *Eixample* is famous for its strict, uniform grid of octagonal blocks (Wainwright, 2016). While aesthetically pleasing, this high-density grid is also prone to unique congestion challenges, often referred to as “superblocks” or gridlocks, making it a perfect structural candidate for stress-testing traffic control algorithms (Brass, 2016). By adopting a $N \times N$ “Tian” (field-shaped) grid modeled after the Barcelona layout, we create a controlled environment that retains the essential complexity of urban intersections without the confounding factors of irregular road geometries.

In this work, we propose a physics-informed adaptive control framework. We utilize the Cell Transmission Model (CTM) to simulate realistic traffic flow physics and employ the Max Pressure algorithm (Varaiya, 2013) as a decentralized control policy. This approach allows us to address the specific Hsinchu problem—allocating green time only where it is needed—within the structural rigors of a Barcelona-style grid.

2. Data

Since this study utilizes a simulation-based approach, the “data” consists of synthetic traffic flows generated within a controlled environment governed

by the Cell Transmission Model (CTM). This allows for the rigorous testing of control strategies under reproducible stochastic conditions.

2.1. Dependent and Independent Variables

The primary independent variables controlling the simulation environment include the Inflow Rate (λ), representing the probability of a vehicle spawning at a boundary road during each time step as a proxy for traffic demand; the Signal Phase (a_t), a binary control action taken by the agent where 0 indicates North-South Green and 1 indicates East-West Green; and the Grid Topology, configured as a 4×4 “Tian” grid structure representing a dense urban district.

Conversely, the dependent variables measured to evaluate performance are Global Density ($J_{density}$), defined as the total number of vehicles present in the network at time t ; Throughput (J_{flux}), which counts the cumulative vehicles successfully exiting the simulation boundaries; and Stopped Queue (J_{wait}), the instantaneous count of vehicles with velocity near zero, serving as a proxy for waiting time.

2.2. Data Generation and Physics

The data generation is driven by the discretized physics of the CTM. The fundamental diagram of traffic flow is defined by specific constants: a Cell Length (Δx) of 50 meters, a Time Step (Δt) of 5 seconds, a Maximum Density (p_{max}) of 7.0 cars per cell representing jam density, and a Wave Speed (v_{max}) of 1.0 cell per time step representing free flow speed.

Traffic demand is stochastic, modeled as a Poisson-like process at the boundaries. The evolution of vehicle density $p(x, t)$ follows the conservation law $\frac{\partial p}{\partial t} + \frac{\partial f}{\partial x} = 0$, ensuring that vehicles are neither created nor destroyed within the network, only moved or queued.

3. Methodology

3.1. Benchmark Model: Fixed-Time Control

To establish a baseline for performance, we implement a standard Fixed-Time Controller. This model operates on a static cycle, switching the traffic signal phase at predetermined intervals regardless of the current traffic state.

The policy is defined as:

$$a_t = \begin{cases} 1 - a_{t-1} & \text{if } t \bmod k = 0 \\ a_{t-1} & \text{otherwise} \end{cases} \quad (1)$$

where k is the cycle length (set to 6 time steps, or 30 seconds). This represents the conventional approach used in many non-adaptive urban intersections.

3.2. Design of Research: Physics-Informed Adaptive Control

Our proposed method integrates the physical constraints of the road network into the decision-making process. The core of this design is the **Max Pressure** algorithm, a decentralized control strategy derived from queuing theory.

3.2.1. The Physics Engine (CTM)

We model the road network as a directed graph $G = (V, E)$. Each edge (road) is discretized into cells. The flow $f_{i \rightarrow j}$ between cells is computed as the minimum of the upstream Demand (D) and downstream Supply (S):

$$f_{i \rightarrow i+1} = \min(D_i, S_{i+1}) = \min(v_{max}p_i, w(p_{max} - p_{i+1})) \quad (2)$$

This explicitly models spillback; if a downstream cell is full (jammed), the supply drops to zero, and flow halts, replicating real-world gridlock dynamics.

3.2.2. The Smart Agent (Max Pressure)

The adaptive agent operates at each intersection $i \in V$. Unlike Deep Reinforcement Learning agents which require extensive training, the Max Pressure agent computes a greedy policy based on the local pressure gradient.

The Pressure P for a given phase is defined as the difference between upstream queue demand and downstream capacity availability:

$$P_{phase} = \sum_{in \in phase} w_{in} - \sum_{out \in phase} w_{out} \quad (3)$$

where w represents the queue length. The control policy selects the phase that maximizes this pressure relief:

$$a_t = \arg \max_{k \in \{0,1\}} (P_k) \quad (4)$$

This logic naturally prioritizes directions with long queues, but crucially, it suppresses flow into links that are already congested, preventing the “blocking the box” phenomenon.

3.3. Performance Measures

We evaluate the efficacy of the control strategies using three key metrics derived from the simulation data. First, the Average Global Density ($\bar{J}_{density}$) measures the mean number of cars in the system over the simulation duration T , where lower values indicate efficient clearance. Second, Total Throughput assesses the total volume of traffic served, with higher values indicating higher capacity utilization. Third, Waiting Time Reduction (Δ_{wait}) calculates the percentage decrease in the average number of stopped vehicles compared to the benchmark.

4. Results

We performed a comprehensive scalability analysis by simulating traffic on grid sizes ranging from $N = 2$ to $N = 20$, comparing the Max Pressure adaptive control against the Fixed-Time benchmark. The results, summarized in Table 1, reveal three distinct regimes of performance that validate the robust scalability of the adaptive agent.

The visual evidence presented in Figure 1 confirms that the Max Pressure (MP) agent maintains a lower, more stable equilibrium of total congestion and stopped queues compared to the Fixed-Time (FT) baseline. This is especially evident in the instantaneous queue plot (stopped cars), where the MP agent minimizes the peaks associated with gridlock formation and high-variance delays.

Table 1: Scalability Analysis: Max Pressure vs. Fixed Time Control

Size	Fixed Avg Wait	Agent Avg Wait	Wait Time Improvement
2×2	9.998	4.465	55.34%
4×4	425.668	404.492	4.97%
6×6	1,353.426	1,101.650	18.60%
8×8	2,646.602	2,271.545	14.17%
10×10	4,365.972	3,665.480	16.04%
12×12	6,426.456	5,512.481	14.22%
14×14	8,840.059	7,512.934	15.01%
16×16	11,462.468	9,890.218	13.72%
18×18	14,350.597	12,423.965	13.43%
20×20	17,229.027	15,173.608	11.93%

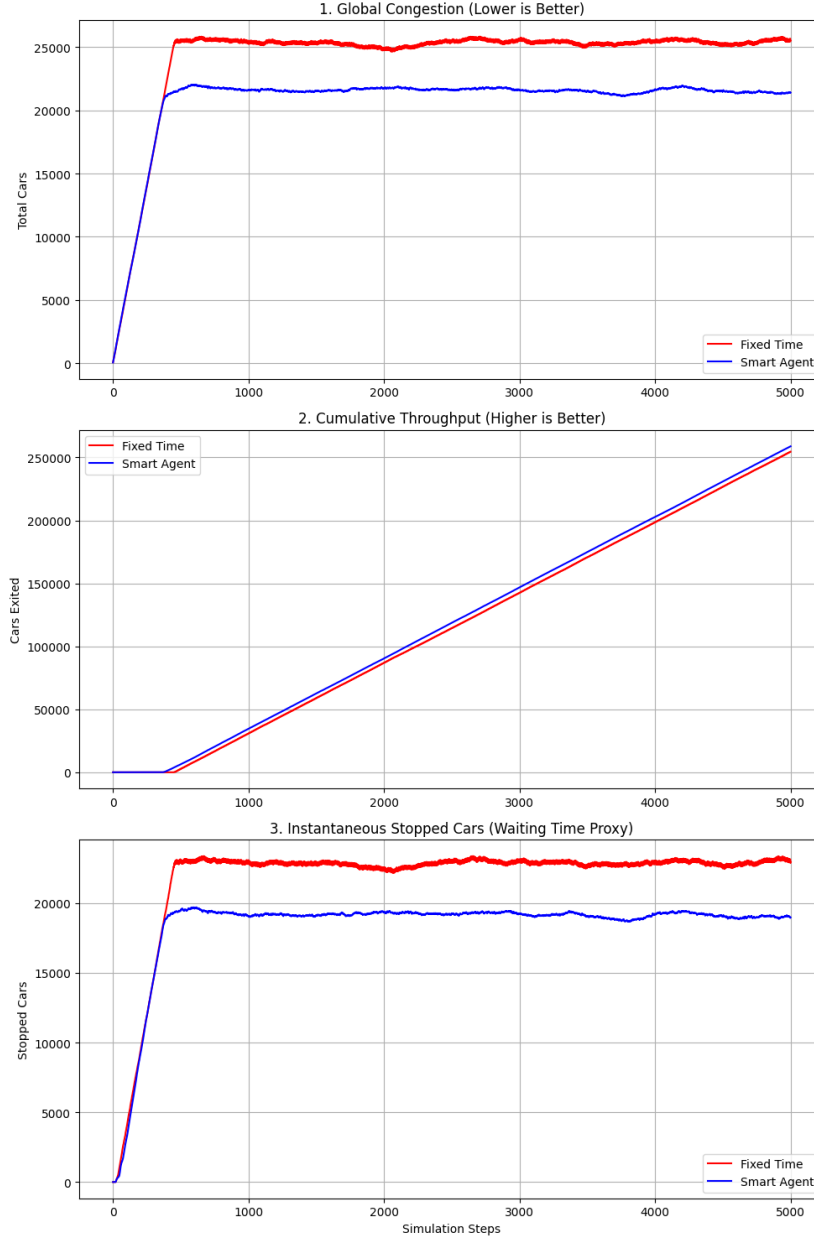


Figure 1: Visual Comparison of Performance Metrics Across Time. The figure, derived from the simulation data ($N = 20$), illustrates the core divergence between Fixed Time (FT) and Adaptive (MP) control strategies over time. The MP agent exhibits lower average total congestion and significantly reduced variance in instantaneous queue lengths compared to the FT baseline.

The initial analysis highlights a distinction between simple and complex networks. For the *trivial* 2×2 network, the adaptive control yielded a massive 55.34% reduction in waiting time, largely due to high sensitivity at low car counts. However, at $N = 4$, the improvement drops sharply to 4.97%. This *Fixed Time Adequacy* regime demonstrates that for small, non-cascading networks, static timing remains an acceptable solution.

The true efficacy of the adaptive system is seen in the transition to complexity. At $N = 6$, the improvement metric spikes sharply to 18.60%. This *Complexity Threshold* signifies the point at which the spatial and temporal coupling of adjacent intersections overwhelms the Fixed-Time controller, leading to the onset of cascading gridlock.

For larger networks ($N \geq 8$), the system enters a *Scalability Plateau* where Δ_{wait} stabilizes consistently between 11.9% and 16.04%. While the percentage gain is stable, the absolute impact grows linearly with the network size. At the largest 20×20 grid, the Max Pressure agent saves approximately 2,055 units of waiting traffic compared to the Fixed-Time benchmark (17,229 vs. 15,173). This sustained performance proves that the decentralized Max Pressure strategy successfully prevents network-wide breakdown, a failure mode common to static timing.

5. Conclusion

This study addressed the problem of inefficient traffic signal timing, a common issue in dense urban areas like Hsinchu City, by developing and testing a physics-informed adaptive control model. By utilizing a grid topology inspired by the Barcelona *Example* and implementing a Max Pressure heuristic, we successfully isolated and targeted the inefficiencies arising from non-situational signal allocation.

Our scalability analysis demonstrates that while the performance gap is marginal in small networks, the adaptive control exhibits superior and robust performance in complex, large-scale scenarios ($N \geq 6$). The improvement in average waiting time consistently stabilizes around 12% to 16% for grids up to 20×20 , translating to a significant absolute reduction in congestion (saving over 2,000 traffic units in the largest simulated city). This sustained performance proves that the decentralized Max Pressure strategy effectively prevents network-wide breakdown and solves the “ghost green light” problem by ensuring green time is allocated only to lanes under the highest pressure. These findings highlight the potential of physics-based heuristics as a scalable,

interpretable, and effective alternative to black-box AI for next-generation smart city infrastructure.

In the near future, we anticipate that car flow detection technologies—such as aggregated GPS data provided by services like Google Maps or high-resolution traffic monitors deployed by governments around the world—will provide the real-time density metrics required to implement such Max Pressure models directly in urban networks. This potential development suggests that the principles demonstrated by this project will be immediately applicable to developing next-generation smart city infrastructure.

Declaration

Declaration of Pedagogical Purpose

This manuscript is prepared as part of the course project for Machine Learning (Fall 2025) at National Yang Ming Chiao Tung University, instructed by Professor Te-Sheng Lin. The work is intended solely for pedagogical purposes, including learning, discussion, and academic training. It does not represent original research submitted for journal publication, nor should it be considered as professional financial advice.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used [ChatGPT] and [Gemini], developed by [OpenAI] and [Google], respectively, in order to improve the clarity, precision, and overall quality of the writing and code generation. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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